MILITARY SPECIFICATION

CONTROL-MONITOR SET, GUIDED WEAPON, AN/AWW-7D, GENERAL SPECIFICATION FOR

This specification is approved for use within the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the Guided Weapon Control-Monitor Set AN/AWW-7D herein referred to as the pod. The pod is for airborne use and performs as a major component part of the data link communication network, providing transmission of command signals and reception of video data, between a control aircraft and a Walleye guided weapon.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 93, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-W-5088	Wiring, Aircraft, Selection and Installation of
MIL-E-5400	Electronic Equipment, Aerospace General Specification for
MIL-T-5422	Testing, Environmental, Airborne Electronic Equipment
MIL-M-7793/6	Meter, Time Totalizing
MIL-A-8591	Airborne Stores, Associated Suspension Lugs and Aircraft Stores Interface, General Design Specification for
MIL-E-17555	Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-H-46855	Human Engineering Requirements for Military System Equipment and Facilities
MIL-T-81624	Television Display Requirements for the Weapon System for Guided Weapon Mark 1 Mod 0 and Mod 2 (Walleye)

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

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MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-965	Parts Control Program
MIL-STD-1472	Human Engineering Design Criteria for Military Systems Equipment and Facilities

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

DL 875AS1000 and all drawings therein	Drawing List for Guided Weapon Control Monitor Set AN/AWW-7D

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Parts and materials.

3.2.1 General. Parts and materials used in the control-monitor sets shall conform to the requirements of the drawings and subsidiary specifications listed in the drawing list (DL) specified herein except, insofar as errors on the drawings may be incompatible with the requirements of this specification and related specifications. Unless otherwise specified in the contract, proposed changes, either to correct drawing discrepancies or to effect improvements, shall be subject to the acceptance by the procuring activity.

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3.2.1.1 Nonstandard parts and material approval. Approval for the use of nonstandard parts and materials shall be obtained by the contractor as outlined in MIL-E-5400, and consistent with procedures cited in MIL-STD-965.

3.2.1.2 Basic conformance. Parts and materials selected for use in the pod shall be in accordance with MIL-E-5400.

3.3 Design and construction. The control-monitor set shall be manufactured in accordance with the drawings and subsidiary specifications listed in DL 875AS1000, except insofar as errors of omission or commission on the drawings may be incompatible with the requirements of this specification. In any case, it is a condition of final acceptance that the pods manufactured shall meet all the performance requirements of this specification and the appropriate test documents. Proposed changes, either to correct drawing discrepancies or to effect improvements, shall be acceptable to the procuring activity prior to their incorporation.

3.3.1 Physical characteristics.

3.3.1.1 Description. The pod shall consist of three major sections: The forward radome assembly, the aft radome assembly, and the casing (center body) assembly. All principle dimensions and locations of particular features, including center-of-gravity, shall be as shown on figure 1 and as specified in table I.

3.3.1.2 Total weight. Total weight of the pod shall be not greater than 645 pounds (292 kilograms).

3.3.1.3 Flight requirements. The Control-Monitor Set, Guided Weapon AN/AWW-7D shall be in accordance with DL 875AS1000 and shall be functionally compatible and flight certifiable as an emergency jettisonable pod on the following aircraft:

a. A-4

b. A-7A/B/E

c. TA-7C

d. F-4

The pod shall be designed for carriage on the centerline or inboard wing-station, as applicable to the specified aircraft, and shall be capable of captive flight.

3.3.2 Reliability.

3.3.2.1 Operational stability. The pod shall operate with required performance, continuously or intermittently, for a period of at least 100 hours or 1 year without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The equipment shall have a total operating life of not less than 300 hours with minimal servicing and replacement of parts.

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NOTES:
1. STATION (STA) INDICATES DISTANCE IN INCHES FROM NOSE OF FORWARD RADOME.
2. UMBILICAL CONNECTION PIN ASSIGNMENT GIVEN IN TABLE II.
3. WEIGHT OF SET IS 645 LBS. (292 KG).
4. VOLUME OF SET IS 10.3 CU. FT. (.292 CU. M.).

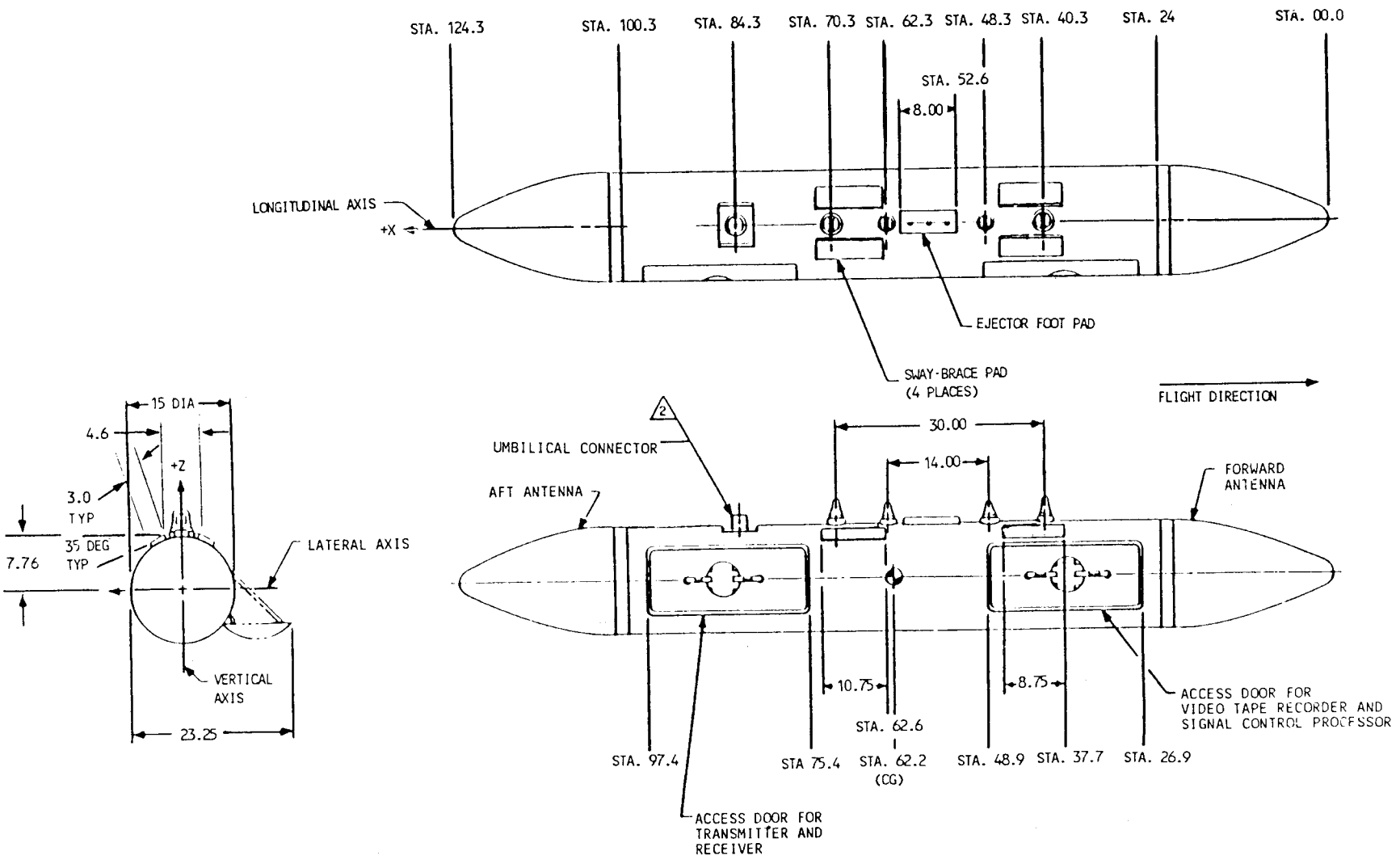


FIGURE 1. Control-Monitor Set, Guided Weapon, AN/AWW-7D.

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TABLE I. Weight and dimensions for the pod.

Parameters	Value
Length	124.3 inches (3.16 meters)
Diameter 1/	15 inches (38.1 centimeters)
Volume	10.3 cubic feet (0.292 cubic meters)
Weight	645 pounds (292 kilograms) typical

1/ Major diameter of casing excluding bomb lug wells, umbilical connector, ejector footpad and sway brace pads.

3.3.2.3 Reliability in mean-time-between-failure (MTBF). The pod shall have a specified mean (operating) time between failures (MTBF) of 50 hours when tested and accepted as specified in 4.5.3.

3.3.2.4 Time totalizing meter. The following units shall contain time totalizing meters in accordance with MIL-M-7793:

Unit	Type Meter
a. Signal Control Processor 875AS1006	M 7793/6-001
b. Radio Transmitter 875AS1008	M 7793/6-001

3.3.3 Cabling and connections.

3.3.3.1 Umbilical connector. Input/output terminations to the pod for primary power, control signals, and audio/video information shall be made through the umbilical connector, A1W1J1, mounted on top of the casing assembly. The connector shall be type M81582/1E03SA and shall mate with the aircraft umbilical connector type M81582/02E03PA1. The umbilical connector pin number assignment, function, and current/voltage requirements shall be as specified in table II. A receptacle cover assembly in accordance with Drawing 2300287 shall be provided to protect the umbilical connector when the pod is not in use. The main cable assembly, terminating in the umbilical connector, shall run the length of the casing assembly and interconnect the main component assemblies.

3.3.4 Controls.

3.3.4.1 Internal adjustments. No adjustments to the pod shall be required prior to its use with the following exceptions: The 1A/C-2A/C switch must be preset to the desired position; the A4/A7-F4 switch must be selected for the correct type of aircraft; the circuit breaker, CB1 must be placed in the ON position; the channel select switch on the transmitter assembly or cockpit channel select switch must be set to match the channel of the weapon; and the video tape recorder must be loaded with tape and placed in RECORD.

3.3.5 Interchangeability. Physical and functional interchangeability shall exist for all units and replaceable assemblies, subassemblies, and parts of Guided Weapon Control-Monitor Set, AN/AWW-7D in accordance with

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MIL-E-5400. DL 875AS1000 shall be used as the standard to measure interchangeability for the pod.

3.3.6 Interference control. The generation of electromagnetic interference by the pod and the vulnerability of the pod to electromagnetic interference shall be controlled within the limits of MIL-STD-461 for Class A1 and Class B1 equipment. EMI testing shall be accomplished under standard conditions specified in 3.3.13 and in accordance with the procedures of MIL-STD-462 to demonstrate compliance with the requirements of MIL-STD-461. An EMI/EMC test plan and a test report shall be prepared.

3.3.7 Maintainability. Maintainability provisions shall be incorporated within the design to provide for service, access, and repair of all assemblies. The replacement of major components within the pod shall be accomplished without soldering and without the need for special tools. Normal routine testing of the pod shall be accomplished using the special test equipment listed in the operation and maintenance instructions manual for the pod, in addition to the standard test equipment normally available in Intermediate Maintenance shops.

3.3.7.1 Test connections. Normal testing of the pod shall be accomplished without disassembly except that the forward and aft antenna terminations shall be disconnected so that RF test equipment may either monitor or supply signals to the antenna feed cables.

3.3.7.2 Built-in-test. The major components that compose the pod shall have built-in-test (BIT) features to check critical parameters.

3.3.7.3 Periodic maintenance. The pod shall require one year periodic maintenance with the exception of the following:

a. The desiccant breather system and desiccant storage bags shall require periodic servicing.

b. The video tape recorder shall require periodic cleaning of the recording heads and tape path to ensure that recordings meet the specified requirements.

3.3.8 Transportability. The Guided Weapon Control-Monitor Set AN/AWW-7D shall be transportable by air, rail, truck and ship when mounted in a shipping container as specified in section 5.

3.3.9 Storage life.

3.3.9.1 Packaged condition. The pod shall be capable of being stored in unsheltered areas when packaged to the requirements of section 5. The normal storage environmental extremes shall be at any relative humidity between -65°C and 71°C.

3.3.9.2 Unpackaged condition. The pod shall be capable of safe storage in the unpackaged, assembled condition for not less than 2 years, in the weapon ready storage area of a shore installation, without impairment to form, fit, or function. Normal storage environment extremes shall be at any relative humidity from -54°C to +60°C. Covers and plugs shall be in place during storage. The desiccant breather system and desiccant storage bags shall be

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serviced, replaced, or tested periodically as specified in the operation and maintenance instructions manual for the pod. No realignment of the pod, or its component parts shall be necessary after storage.

3.3.10 Nameplates and identification marking. Serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307.

3.3.10.1 Decals, nameplates, and marking locations. All decals, nameplates, and marking shall be affixed to the pod in accordance with DL 875AS1000, and all referenced marking specifications, documents, and drawings thereon.

3.3.10.2 Serial numbers. Serial numbers will be assigned by the procuring activity for the pods under contract or order. The contractor shall serialize sequentially all pods which are delivered under the contract.

3.3.10.3 Special warning decals. The forward and aft radome assemblies shall be marked on each side surface and upper surface with warning decals labeled "NO STEP RADOME". The decals shall meet all requirements of Drawing 184AS246.

3.3.11 Safety. The contractor shall ensure that system safety requirements are in accordance with MIL-E-5400.

3.3.11.1 Flight safety. Flight safety procedures applicable to the associated aircraft and the Walleye guided weapon shall be applicable to the Guided Weapon Control-Monitor AN/AWW-7D.

3.3.11.2 Ground safety. Ground safety procedures applicable to the associated aircraft and the Walleye guided weapon shall be applicable to the Guided Weapon Control-Monitor AN/AWW-7D.

3.3.12 Human performance and human engineering. The pod and all component parts thereof, shall be designed in accordance with the requirements of MIL-STD-1472. Recognized principles of human engineering as specified in MIL-H-46855 shall be followed during design.

3.3.13 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient ($25^{o} \pm 5^{o}C$)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative
e.	Input power voltage	115 ± 3.0 Vrms, 400 ± 10 Hz, 3-phase and 28.0 ± 2.0 vdc.

3.3.14 Service conditions. The pod shall operate as required under any of the environmental service conditions or reasonable combinations of these

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conditions as specified in MIL-E-5400 for Class I equipment, except as modified herein.

3.3.14.1 Vibration. The equipment shall meet the performance requirements of this specification during and after being subjected to the vibration environment specified in 4.5.6.1. The performance characteristics specified in 3.4 shall apply.

3.3.14.2 Temperature-altitude. The pod shall meet the temperature-altitude combination requirements of MIL-E-5400 for Class 1 equipment except the maximum altitude shall be 55,000 feet (see 4.5.6.2).

3.3.14.3 Humidity. The pod shall operate as required after being subjected to a humidity test. The pod shall not be damaged as a result of the humidity test specified herein (see 4.5.6.3).

3.3.14.4 Shock. The pod shall meet the performance requirements of this specification after being subjected to a shock test (see 4.5.6.4).

3.3.14.5 Sand and dust. Sand and dust requirements shall be as specified in MIL-E-5400 (see 4.5.6.5).

3.3.14.6 Fungus. Fungus requirements shall be as specified in MIL-E-5400 (see 4.5.6.6).

3.3.14.7 Salt atmosphere. Salt atmosphere requirements shall be as specified in MIL-E-5400 (see 4.5.6.7).

3.3.14.8 Explosive conditions. Explosive conditions requirements shall be as specified in MIL-E-5400 (see 4.5.6.8).

3.3.14.9 Rain. The equipment shall operate as required after exposure to rainfall conditions (see 4.5.6.9).

3.3.14.10 Acceleration. The pod shall operate as required while being exposed to the design requirements of MIL-A-8591 for both wing and fuselage mounting. The pod shall withstand the aerodynamic and inertial captive flight loads specified in MIL-A-8591 for flight regimes up to MACH .95 at sea level (see 4.5.6.10).

3.3.15 Environmental control. Environmental control shall consist of natural circulation of air within the pod enclosure for dissipating heat generated by the electronic components. Heat shall also be dissipated by conduction through the component mounting bases and the casing to outside ambient air. The tape recorder shall be protected from temperatures below $0^{o}C$ (nominal) by heaters. Humidity inside the pod enclosure pod shall be controlled by a desiccant breather system. The breather system pressure differential limits shall be not greater than 3 psi.

3.3.16 Warm-up time. The time required for the pod to warm up prior to operations shall be not greater than 5 minutes, either under standard conditions or at extreme service conditions, except for the recorder (see 3.7.5.1.1).

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3.3.17 Input electrical power.

3.3.17.1 Operating power. The equipment shall meet all applicable requirements of MIL-STD-704 and shall perform as required when energized from the following power sources having characteristics and limits as specified in MIL-STD-704. The power required shall be not greater than the specified amounts.

a. AC Power (three phase), 115 V, 400 Hz, 1,050 Watts maximum.

b. DC Power, 28 V, 2.6 Watts maximum.

3.3.17.2 Protection. The equipment shall be capable of withstanding input power variations as imposed by abnormal electric system operations as specified in MIL-STD-704. Degraded performance will be permitted for pod operating during the emergency mode of steady state AC power as specified in MIL-STD-704. Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electrical system operation.

3.4 Performance. The performance of the pod shall be as specified herein plus any and all performance requirements stated on the drawings of DL 875AS1000. Unless otherwise specified herein, these requirements shall apply under both standard and service conditions. When reduced performance requirements for service conditions are acceptable, such variations are specified herein.

3.4.1 Operation. The Guided Weapon Control Monitor AN/AWW-7D shall operate from the aircraft power system. The pod shall receive data link multiplex signals from the aircraft cockpit controls (see table II of Aircraft/Pod Interface), and provide to the cockpit monitor the received weapon video, including signals from the weapon acknowledging launch and target tracking, superimposing them onto the received video signal. The transmitter and receiver carrier frequencies shall be selected between five different frequencies as specified in the contract (see 6.2.1).

3.4.2 Functional organization. During operation of the pod, the operator shall have the capability of selecting, from the cockpit, either a forward or aft data link pod antenna. A diplexer 196AS298 shall be used to provide isolation between the command transmitter output and the video receiver input to prevent receiver desensitization, allowing utilization of one antenna for simultaneous transmission and reception of signals. Video data link signals, received at the selected antenna, shall be filtered by the diplexer and applied to the video receiver. The receiver shall decode the received video and relay it to the aircraft cockpit video monitor. The data link pod command transmitter shall accept analog weapon commands from the cockpit controller. The transmitter shall modulate these analog signals onto the command RF carrier and transmit them to the weapon.

3.5 Schematic diagram. The electrical interconnection of components comprising the pod shall conform to Drawing 875AS1003.

3.6 Interface description.

3.6.1 Electrical interface. All communication between the data link pod and the control aircraft cockpit shall be accomplished utilizing an umbilical cable connecting the pod to the aircraft. The electrical signal characteristics for aircraft/pod interface shall be as specified in table II. Multiple

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frequency tones originating in the cockpit controller are used as the pod control signals, and are applied to the pod through the aircraft wiring. These various tones shall order the pod to initiate, and transmit, specific command signals to the weapon. Table III includes the specific functions, tone frequencies, and driving currents for the multiplex control signals. The pod channels shall be selected as denoted on the CHANNEL SELECT switch mounted on the transmitter assembly in the pod or as denoted on the COCKPIT CHANNEL switch of the aircraft. The pod shall operate from the aircraft power system.

TABLE II. Aircraft/Pod interface.

AN/AWW-7D umbilical connector (M81582/1E03SA) pin number assignment, function and current/voltage requirements		
Connect Pin No.	Function	Current/Voltage
1	Ground	
2	N/A	
3	Multiplex Control (Signal, Current Inputs) A4/A7	60 Milliamperes (mA) Peak to Peak 25 mA RMS Maximum
4	Ground	
5	Pod Status Indicator (PSI) F4	+28 Vdc, 50 mA Maximum
6	N/A	
7	N/A	
8	Trigger	+28 Vdc, 22.8 mA Maximum
9	Pod Status Indicator (PSI) A4/A7	+28 Vdc, 50 mA Maximum
10	N/A	
11	Station Select	+28 Vdc, 100 mA Maximum
12	N/A	
13	Frequency Select	+28 Vdc, 115 mA Maximum
14	Ground (WPN I.D. GND) Station	Required for A/C to Operate
15	Audio Input (To VTR)	10 Vac p-p Maximum Signal into 620 Ohms
16	Frequency Select	+28 Vdc, 115 mA Maximum
17	N/A	
18	Cockpit Video Coax Shld	
19	Cockpit Video Coax Shld	
20	Ground	
21	N/A	
22	Multiplex Control (Signal, Current Inputs) F4	60 Milliamperes (mA) Peak to Peak 25 mA RMS Maximum
23	Ground	
24	N/A	
25	N/A	
26	N/A	
27	115Vac, 400 Hz, Ø A Input	3 Amperes Maximum
28	115Vac, 400 Hz, Ø B Input	3 Amperes Maximum
29	115Vac, 400 Hz, Ø C Input	3 Amperes Maximum
30	3 Ø Neutral Input	3 Amperes Maximum
31	N/A	

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TABLE II. Aircraft/Pod interface - Continued.

AN/AWW-7D umbilical connector (M81582/1E03SA) pin number assignment, function and current/voltage requirements		
Connect Pin No.	Function	Current/Voltage
32	Power Change Over (PCO) Indicator	A Ground in AN/AWW-7D Electronics, 1 Ampere Maximum
33	N/A	
34	N/A	
35	Practice Guided WPN VTR Control	
36	Cockpit Video Coax Shld	
37	Cockpit Video	3 Volts p-p Composite Video per MIL-R-81624 into 91 ohms
38	Cockpit Video Coax Shld	
39	N/A	
40	N/A	
41	Multiplex Signal Shld	
Pin 16 Energized - Activates Frequency Channel 1 13 Energized - Activates Frequency Channel 2 13 + 16 Energized - Activates Frequency Channel 3		

3.6.2 Mechanical interface. The mechanical interface between the pod and aircraft shall ensure compatibility with all Walleye configured bomb-racks, including AERO 7A, AERO 20A, MAU-9A/A, MAU-12B/A and BRU-10. The umbilical connector, bomb lug wells and suspension lugs, and ejector foot and sway brace pads, shall be dimensioned and located as shown on figure 1. Both 14- and 30-inch spacings shall be available on the pod. The ejector footpad shall engage the stores-away switch in the aircraft bomb rack when the pod is loaded onto the rack.

3.6.3 Pod/test equipment interface. The input power for flight line or shop test shall be applied to the pod J1 umbilical connector via the Distribution Box J3425A/AWW, which shall also control the input signals from the tones generated by the audio generator. The internal components shall be easily accessible for test by opening either one or both of the two side doors. The pod J1 umbilical connector inputs during test shall be as specified in table II.

3.6.4 General arrangement. The pod shall be designed for suspension from a standard Walleye-configured bomb rack, and for non-jettisonable carriage on either the center line or inboard stores stations for a variety of high performance aircraft.

3.7 Detail requirements.

3.7.1 Functional elements. The pod functional elements shall consist of the following:

a. Transmitter, Radio T-13660A/AWW-7.

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b. Control-Processor, Signal C-10292A/AWW-7.

c. Receiver, Television R-2066A/AWW-7.

d. Recorder, Tape RO-499A/AWW-7.

e. Antenna-Radome AS-3220/AWW-7 (Forward/Aft).

TABLE III. Aircraft/pod functions and tone frequencies.

Tones (KHz)	Currents (mA)	Functions
1.3 ± 1.42%	.10 ± .015 mA	Antenna Select
1.7 ± .94%	.10 ± .015 mA	Test
2.3 ± 3.0%		Pod Status Indicator
5.4 ± .81%	.12 ± .02 mA	Video ON
7.35 ± .73%	.175 ± .02 mA	Video OFF
10.5 ± .75%	.27 ± .025 mA	Command Override
22 ± 8.18%	.70 ± .04 mA	Azimuth
30 ± 8.15%	1.17 ± .06 mA	Elevation

3.7.2 Transmitter, Radio T-1366A/AWW-7. The transmitter assembly, A3, shall consist of the following major components:

a. Transmitter, command, TR1.

b. Diplexer, Z1.

c. Power supply assembly, VR1.

d. Switch, rotary (channel select), S1.

e. Switch (1A/C - 2A/C select), S2.

f. Switch (aft antenna test), S3.

g. Switch (A4/A7 - F4 select), S4.

h. Switch, coaxial (attenuator), K1.

i. Switch, coaxial (antenna select), K2, K3, K4.

j. BIT board assembly, TB2.

k. Oscillator, Y1.

l. Mixer, Z2.

m. Terminal board, A1, (Frequency Select).

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3.7.2.1 Transmitter, command. The command transmitter, TR1, shall accept analog signals, modulate these signals onto the command RF carrier and transmit them via the antenna. The transmitter shall exhibit the following performance requirements:

a. The command transmitter shall meet the specifications of Drawing 196AS238.

b. Power output of the transmitter shall be not less than 37.5 dBm into 50 ohms when measured at either antenna termination.

c. The transmitter output shall be enabled and disabled by command of the aircraft cockpit controls.

d. A command receiver, set to receive the signal transmitted by the pod, shall demodulate the carrier and present signals which demonstrate the characteristics of the original weapon command signals from the cockpit controller. These characteristics shall include frequency stability of the control tones within the defined limits, harmonic distortion limited to 2 percent of the signal, cross talk between signals not less than 6 dB below the desired signal, and linear deviation of the Azimuth and Elevation Control tones with respect to the oscillator control voltages over their bandwidth.

3.7.2.2 Diplexer. The diplexer, Z1, shall permit one antenna to be used for transmission and reception of RF signals simultaneously by combining the transmitter and receiver bandpass responses to a common response at the antenna. Conversely, the diplexer shall provide isolation between the command transmitter output and the video receiver input to prevent interference while operating on the common antenna. The diplexer shall be in accordance with Drawing 196AS298.

3.7.2.3 Power supply assembly. The transmitter power supply, VR1, Drawing 196AS180, shall provide 28 volts dc (+1.5/-1.0 Vdc) at up to 3 amperes current to power the transmitter assembly. The power supply shall operate from the aircraft 115 Vac, 400 Hz, 3-phase power source with transformer T1, Drawing 184AS300, providing the voltage conversions required by the regulator circuitry. The load regulation of the power supply shall be ± 1 percent or better. Ripple voltage shall be not greater than 50 mv peak-to-peak under no load conditions or 100 mv pp under transmitter power output conditions; transients and voltage spikes shall be not greater than 150 mv under no load, 500 mv under normal load, or 500 mv under maximum load conditions. The power supply shall be protected against overload and short circuit current shall be not greater than 0.5 ampere. The power supply shall return to normal operation when the overload condition is removed.

3.7.2.3.1 BIT board assembly. The BIT board assembly, TB2, Drawing 1035AS150, shall contain LED indicators which signal "normal" or "fault" operation of the power supply by detecting the output voltage and shall be attached to the power supply assembly. In addition, an LED shall indicate when the command transmitter output power is 5 watts or greater. A final LED shall indicate when the BIT oscillator, Y1, is activated.

3.7.2.3.2 Transmit relays. The command to transmitter power output control relay, VR1K2, and attenuator switch control relay, VR1K1, shall be

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incorporated in the power supply assembly. At the functional command of the cockpit control, an electrical ground shall be applied to the unterminated return wire of the two relay coils. Relay VR1K2 shall then apply 28 volts DC to the transmit supply pin on the command transmitter. Relay VR1K1 shall remove the ground termination from coaxial switchable attenuator A3K1 which acts to remove the 30 dB attenuation from the transmitter power output circuit.

3.7.2.4 Attenuator switch. The coaxial switchable attenuator A3K1, shall be a switchable 30 dB coaxial attenuator which permits low level power transmission from the transmitter during a pod/weapon interface test on one aircraft. It shall also attenuate the transmitter output signal during BIT operation of the pod. The attenuator shall be bypassed (switched out of the circuit) during command transmissions from the pod to the weapon.

3.7.2.5 1 A/C - 2 A/C select switch. Switch S2 shall be a lever lock switch which selects between two operating modes of the transmitter assembly during test. In the 2 A/C mode, the 30 dB switchable attenuator, A3K1, shall be under control of the attenuator switch control relay VR1K1. In the 1 A/C mode, the 30 dB attenuator shall remain in the transmitter power output circuit when the aircraft cockpit control is in the test mode. In any other cockpit control mode, the attenuator switch shall be under control of relay VR1K1.

3.7.2.6 Channel select switch. The channel select switch, S1, shall permit the command transmitter carrier frequency and video receiver channel to be selected between 5 different channels (1, 4, 2, 5, 3). The terminal board A1 shall select the RF frequencies 1, 2, or 3 from the cockpit when digital signals are applied as specified in table II. Switch S4 selects A4/A7 or F4 aircraft operation.

3.7.2.7 Forward antenna test switch. The forward antenna test switch, S3, shall be a push-button momentary-on switch which permits an operator to manually select the forward antenna during pre-flight operational test of the pod. Indicator lamp DS2, located on the Signal Control Processor front panel, shall be illuminated when this switch is in the ON position.

3.7.2.8 Antenna select switches. Normally, the telemetry signals shall be transmitted and video signals received by the aft antenna. At the command of the cockpit controller however, the forward antennas may be selected for communication between the pod and weapon. The antenna select switches, K2, K3, and K4, shall be coaxial remote control switched, which provide this capability. In addition, they shall enable mounting of RF signals internal to the equipment during BIT operation of the circuitry. The insertion loss of each switch shall be not greater than 0.3 dB.

3.7.2.9 BIT oscillator. Oscillator Y1 shall be a fixed tuned RF oscillator which meets the requirements of Drawing 1035AS194. It shall be activated during BIT operation of the pod. Its output signal shall be mixed with the transmitter RF carrier to provide a BIT video signal for the receiver. The oscillator frequency will be specified in the contract (see 6.2.1).

3.7.2.10 Mixer. Mixer Z2 shall combine the BIT oscillator signal and the command transmitter RF carrier channel 2 to provide a BIT video signal for the receiver. The mixer shall meet the requirements of Drawing 1035AS193.

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3.7.3 Signal Control-Processor C-10292B/AWW-7. The signal control processor, A2, shall be manufactured in accordance with Drawing 875AS1006 and shall meet the requirements of 3.7.5.1 through 3.7.3.7.2.

3.7.3.1 Signal control processor functional description. The signal control processor (SCP) shall process the cockpit control signals and supply the signals to the command transmitter modulation circuit for transmission to the weapon. The SCP shall also process the video output of the receiver and supply video signals to the aircraft cockpit monitor and to the video tape recorder. Performance requirements for the data link SCP are as follows:

a. The SCP shall control local application of aircraft power to the video receiver, and video tape recorder.

b. The SCP shall control the remote selection of the multiplex line.

c. The SCP shall provide regenerated sync and blanking/horizontal drive for a sync comparison test, and provide voltage comparison results for a PLL (Phase Locked Loop) sync regenerator test.

d. The SCP shall provide tone decoding circuits and provide video modulation for the transmitter (only during built-in-tests).

e. The SCP shall remove sync pulses from the composite video signal, and provide vertical sync and horizontal sync pulses for sync regenerator control when the weapon video is received.

f. The SCP shall provide automatic gain controlled video to the video summing amplifier.

g. The SCP shall preamplify the video signal from the receiver and provide automatic gain controlled video for the pre-amplifier. The SCP shall also sum the regenerated sync with the video, and provide a composite video output signal to be displayed on the aircraft cockpit video monitor.

h. The SCP shall supply +28 volts and +12 volts dc for operation of the video tape recorder.

i. The SCP shall provide logic gating for operation of the pod transmitter, recorder, and antenna switch.

j. The SCP shall provide phase lock loop circuitry to decode tones to control various functions of the pod as listed in table IV.

k. The SCP shall receive multiplex current signals, convert them to voltage signals, and add the 2.3 KHz tone for frequency modulating the transmitter.

l. The SCP shall provide oscillator circuit to generate the 2.3 KHz tone and provide a video buffer for the video tape recorder.

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m. The SCP shall contain built-in-test (BIT) circuitry including: logic circuits for receiving BIT start-stop signals and generating GO and NO-GO signals for front panel light emitting diodes (LEDs); BIT data selector and counter for generating test results and for controlling front panel test fault indicators; BIT decoder, relays, and logic gate circuits for controlling when various parameters are to be monitored and/or rerouted during BIT.

3.7.3.2 Signal Control Processor (SCP) content. The SCP contains the following circuit card assemblies:

a. A1 - Phase-Lock Loop Sync Regenerator.

b. A2 - BIT Board No. 1 (BIT Signal Generator).

c. A3 - Sync Stripper/Oscillator Assy.

d. A4 - Sync Remover/Horizontal Generator Assy.

e. A5 - Video Amplifier (including AGC Preamplifier).

f. A6 - Power Supply (Voltage Regulators).

g. A7 - Logic Decoding Interface.

h. A8 - PLL Tone Decoder and Tone Filters.

i. A9 - PLL Tone Decoder and Tone Filters.

j. A10 - Line Receiver and Summing Amplifier.

k. A11 - 2.3 KHz Oscillator and (VTR) Video Driver.

l. A12 - BIT Board No. 2 (BIT Controller).

m. A13 - BIT Board No. 3 (Data Selector and Counter).

n. A17 - BIT Board No. 4 (BIT Sequence).

3.7.3.2.1 Station select. The SCP shall control the remote selection of the command signal multiplex line. When the station select voltage, aircraft +28 volts dc from the cockpit, is applied to the pod and the A4/A7-F4 switch (A3S4) is in the A4/A7 position relay A2K1 is energized connecting the A4/A7 multiplex line to the line receiver board in the SCP. If either the station select voltage is not applied or the A4/A7-F4 switch is in the F4 position, relay A2K1 is not energized and the F4 multiplex line is connected to the line receiver board.

3.7.3.2.2 Line receiver. The multiplex line shall supply a nominal 6.8 milliampere peak-to-peak current signal consisting of the multiplexed, pre-emphasized, sub-carrier oscillator frequencies generated by the cockpit controller. The line receiver shall be a low input impedance amplifier which converts the received signal current to a proportional voltage output. This output shall be summed with the 2.3 KHz oscillator output (see 3.7.3.2.4). The resultant signal shall be applied to the transmitter modulation driver circuit. The driver circuit shall buffer the input signals, set the output

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voltage level, and apply these signals to the transmitter frequency modulation input terminal which presents a high impedance load. The modulation signal level shall be adjusted at the driver to give a transmitter deviation of 240 KHz peak-to-peak with the 1.7 KHz, 7.35 KHz, 10.5 KHz, 22 KHz, and 30 KHz signals applied on the multiplex line.

3.7.3.2.3 Tone detectors and logic decoding. The output signal of the line receiver shall be de-emphasized and detected by phase-locked loop detectors which have been tuned to each sub-carrier oscillator frequency comprising the multiplex signal. The resultant dc voltage from each detector shall be applied to the logic decoding circuitry which shall provide selective gating for operation of the pod transmitter, tape recorder, and antenna switch. Table IV specifies the required control functions.

3.7.3.2.4 Arm logic and 2.3 KHz oscillator. Application of +28 volts dc from the cockpit control to the PSI Control Input in the SCP shall energize 2-pole relay A2A11K1. One set of relay contacts shall apply a control signal to the logic decoding circuit which enables the transmitter power output. A second set of relay contacts shall connect the output of the 2.3 KHz oscillator in the SCP to the summing input of the transmitter modulation driver. The oscillator shall supply a 2300 $\pm$ 7 Hz sine wave with amplitude of 0.65 $\pm$ 0.08 volts peak-to-peak.

3.7.3.2.5 PCO indicator and VTR remote control. Application of the Video R.F. ON command from the cockpit shall energize 2-pole latching relay A2A7K1. One set of relay contacts shall apply a signal ground termination to the PCO indicator wire in the umbilical cable which causes the PCO lamp in the cockpit to illuminate. A second set of relay contacts shall connect +12 volts dc from the SCP power supply to the video tape recorder, enabling it to advance tape and record. The relay shall remain in the "ON" position although the Video R.F. ON command is removed. When the Video R.F. OFF command is applied, the relay shall be reset to the "OFF" position. At this time the PCO lamp shall be extinguished and the VTR shall return to the standby mode.

3.7.3.2.6 Forward Antenna indicator. Application of the forward antenna select command from the cockpit controller shall result in the logic decoding circuit activating antenna select switch A3K2 in the transmitter power supply assembly. Transmission and reception of RF signals shall then occur from the forward antenna-radome rather than the aft antenna-radome. Indicator lamp DS2 on the SCP front panel shall illuminate when the forward antenna is selected.

3.7.3.3 Video processor. The Signal Control Processor shall process the receiver video output signal and supply video signals to the aircraft cockpit monitor and the video tape recorder. The processor shall use the received video information and synthesized pulses generated internally to reconstruct a composite video and video sync signal which has continuous sync and relatively uniform contrast during the presentation. Use of the video processor shall eliminate the most disruptive aspects of a received video signal which may be intermittent, have a wide disparity of signal strength, contain overriding noise and have overlapping or phase shifted sync due to multipath reception.

3.7.3.3.1 Preamplifier. The received video signal shall be applied to the video preamplifier which shall provide an automatic-gain-controlled output signal. The preamplifier input impedance shall be 93 $\pm$ 9 ohms. The output signal strength shall be 1.0 $\pm$ 0.2 volts peak-to-peak. Composite sync recovered from the sync remover circuit shall be compared to a preset gain

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control voltage and applied to a feedback circuit which shall control the preamplifier gain, thus providing the AGC for the received signal.

TABLE IV. Command signal and pod response

Input Signal 1/	Resultant Operation
Station Select (+28 VDC from cockpit)	A4/A7 - F4 PSI and multiplex line select.
Pod Status Indicator (PSI) Control (ITL) (+28 VDC from cockpit)	Enable transmitter power output; applies 2.3 KHz tone to transmitter modulation driver circuit.
Forward Antenna Select	(1.3 KHz) Forward antenna connected to RF circuitry; aft antenna disconnected; DS2 lamp on SCP illuminated.
Test Function 2/	(1.7 KHz) Enable transmitter power output; in low power mode if 1A/C mode is selected in pod (high power in 2A/C mode).
Video RF ON	(5.4 KHz) Enable transmitter power output; +12 VDC from SCP applied to VTR - VTR advances tape; signal ground connected to PCO indicator wire.
Video RF OFF	(7.35 KHz) Enable transmitter power output; VTR to standby mode; ground removed from PCO indicator wire.
Command Override	(10.5 KHz) Enable transmitter power output.
Azimuth and Elevation	(22 and 30 KHz) Analog signal are applied to the transmitter modulation input for transmitting to the weapon.

1/ Application of the command signal or cockpit control voltage to the SCP shall result in a pod response as follows.

2/ When transmitter power output is indicated, transmitter TR1 shall transmit the RF output signal in the high power mode. The only exception shall be when the test function is applied and the 1A/C mode is selected, at which time the transmitter output shall be attenuated by 30 dB. Transmitter power output shall be enabled only so long as the appropriate command signal is applied from the cockpit. At all other times, the transmitter shall be in the standby mode (-30 dBm or less).

3.7.3.3.2 Sync remover. A sync remover circuit shall receive the preamplifier video output and then strip off the composite (horizontal and vertical) sync at a constant level on the signal; that being the video blanking pedestal.

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A reference voltage generated by a gate array timed to sample only the horizontal sync pulses shall clamp the video input at this level. By tracking the black picture level, the video signal output shall have the correct contrast although the input signal strength may vary considerably.

3.7.3.3.3 Sync stripper and oscillator. A sync stripper shall separate the sync from the preamplifier composite video output. The vertical sync pulses shall be separated from the composite and sync provided to the sync regenerator circuit. The horizontal sync pulses shall be used to trigger the synchronizable oscillator which shall provide 15,750 Hz pulses for the sync regenerator. The oscillator shall be adjusted for a free-running frequency of 15,600 Hz with no sync trigger input.

3.7.3.3.4 Sync regenerator. The synchronization (sync) regenerator shall consist of a composite television sync generator which shall be triggered by synchronization pulses derived from the received video signal. A phase-locked-loop crystal controlled oscillator shall lock the horizontal sync to the synchronizable oscillator. The vertical sync shall be reset by the vertical sync output of the sync stripper circuit. During periods of intermittently received video signals or noisy sync, the generator shall continue to provide composite sync to the video amplifier. When useable sync is received, the oscillator shall be in phase with it.

3.7.3.3.5 Video amplifier. The video amplifier shall combine the re-generated composite sync with the gain controlled sync - less video from the sync remover circuit and provide a composite video output signal to be displayed on the aircraft cockpit monitor. The output impedance shall be 93 $\pm$ 9 ohms. The composite video signal output shall conform to MIL-T-81624. Video resolution shall be not less than 200 vertical lines and not less than 7 shades of gray shall be differentiated. The specified video output signal shall be present when the video input signal is attenuated by 10 dB and the input signal-to-noise ratio is not less than 3 dB, although a decrease in contrast to 4 shades of gray and a decrease in resolution to 200 vertical lines shall be acceptable. During periods of reception when the picture information is lost due to noise, a raster shall continue to be displayed on the monitor until the picture resumes. (Both horizontal and vertical sync will be present in the video output during these times.)

3.7.3.3.6 Video Buffer. The video signal output from the video amplifier shall be applied to a buffer amplifier which shall provide the video signal source required by the video tape recorder. The buffer shall consist of a current amplifier which shall drive a 75 ohm line. The peak-to-peak video signal measured across a 75 ohm termination shall be not less than 1 volt and shall be not greater than 1.5 volts.

3.7.3.3.7 Video input/output. BNC coaxial receptacles (jacks) shall be mounted on the SCP front panel and shall provide input/output terminations for the video signals in the SCP. Video signals from the receiver shall be applied to the Video In Connector J3, which shall be wired to the preamplifier. BIT video generated in the SCP shall be accessible from J4. Video signals from the VTR shall be applied to connector J5 which shall be wired to the BIT circuitry responsible for testing the video signal. Connector J6 shall provide access to the cockpit video wire in the umbilical cable. The sync-processed video signal output from the video amplifier shall be available from connector J7. Cable A2W1 shall normally be connected between J6 and J7, connecting the video signal output of the SCP to the cockpit monitor.

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3.7.3.4 Audio. Audio information from the aircraft cockpit shall be applied via the SCP to the tape recorder to be recorded on an audio track accompanying the video recording. A 620 ohm, 1/4 watt resistor shall be connected between the audio input and signal ground to match the impedance of the aircraft intercom system.

3.7.3.5 Power distribution and overload protection. Local application of aircraft power to the SCP voltage regulators, transmitter power supply assembly and receiver shall be controlled by the SCP. The three phase power lines shall be protected against overload current by a circuit breaker which is in accordance with Drawing 184AS284 and has the fast time delay range. The total load current on any phase shall be not greater than 3 amperes. The circuit breaker, CB1, shall be located on the SCP front panel.

3.7.3.6 Voltage regulators. The SCP voltage regulators shall supply the power requirements of the SCP circuitry and the video tape recorder. They shall operate from the secondary windings of transformer A2T1, built in accordance with Drawing 196AS196. The aircraft 115VAC, 400 Hz, 3-phase power source shall supply the exciting power to the transformer primary.

3.7.3.6.1 +28 VDC supply. An unregulated +28 volts dc source, operating from the 3-phase secondary winding of transformer A2T1, shall supply power to the tape recorder heater, the +12 volts dc regulator in the SCP and a reference voltage source in the BIT circuitry. The +28 volts dc source shall provide a nominal 3 amperes current and up to 4 amperes when the recorder is first energized. The ripple voltage on the +28 volts dc bus shall be not greater than 2 volts. Load regulation shall be at maximum ± 1 percent. The +28 volts dc source shall be protected from overload currents in excess of 5 amperes by fuse F1, accessible on the SCP front panel. Indicator lamp DS1 on the SCP front panel shall illuminate when the +28 volts dc source is energized.

3.7.3.6.2 +12 VDC regulator. The +12 volts dc regulator shall operate from the +28 volts dc source and shall supply 1.5 amperes nominal current. (This current shall be included in the current requirement of the +28 volts dc supply.)

3.7.3.6.3 +5 VDC regulator. The +5 VDC regulator shall operate from a single phase secondary winding in the transformer and shall supply a nominal 750 mA current. The logic components in the SCP shall be protected from voltage overloads in excess of 7 volts by zener diode A2CR2 on the +5 volts dc bus line.

3.7.3.6.4 Negative voltages. A -12 VDC regulator, operating from a single phase secondary winding in the transformer, shall supply a nominal 380 mA current. A zener diode regulator, operating from the -12 volts dc bus, shall supply a -6.8 volts dc at a nominal current of 230 mA.

3.7.3.6.5 Regulator performance. The ripple voltage measured at the output of any regulator shall be not greater than 100 mv under full load conditions. The load regulation shall be $\pm$ 1 percent or better.

3.7.3.7 Built-in test. The Signal Control Processor shall contain built-in test (BIT) circuitry, including: logic circuits for receiving BIT execute signals and generating GO and NO-GO signals for front panel LEDs; BIT data selector and counter for generating test results and for controlling front panel test fault indicators; BIT decoder, relays, and logic gate

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circuits for controlling the sequence of signal parameters monitored during the test; and a video signal generator for modulating the transmitter during BIT. Table V lists the functions which shall be tested. BIT shall operate correctly over the temperature range of -54°C to +65°C.

TABLE V. BIT sequence.

Test No.	Test Title	Circuit Tested
0	BIT READY (+28 Volts Test)	SCP 28 VDC Source
1	Voltage Tests (± 12, +5, -6 VDC)	A6 Board
2	1.7 KHz Tone Decoder Test	A7, A9 Board
3	10.5 KHz Tone Decoder Test	A7, A8 Board
4	1.3 KHz Tone Decoder Test	A7, A8 Board
5	5.4 KHz Tone Decoder Test	A7, A9 Board
6	7.35 KHz Tone Decoder	A7, A9 Board
7	2.3 KHz Oscillator Using Tone Decoder	A7, A8, All Boards
8	VTR Video Test	VTR
9	Sync Regenerator Test	A1 Board
10	Loop Test Aft Antenna	Pod
11	Lack of Loop Test	Pod
12	Loop Test Forward Antenna	Pod
13	Spare (Automatic GO)	
14	Video Ampere Test	A5 Board
15	Test Complete (GO/NO-GO)	

3.7.3.7.1 Loop test. BIT step 10, the RF loop test, shall be accomplished as follows. BIT video, generated by the SCP, shall be used to modulate the transmitter. The transmitted signal shall be attenuated 30 dB and mixed with the local oscillator, A3Y1, energized during BIT. The mixed signal shall be further attenuated to -78 dBm and routed via the diplexer to the receiver. The received video signal shall be sent back to the SCP video processor which shall compare the received video to the test video. If the received video horizontal sync is of the same frequency and phase as the test video horizontal sync, the SCP shall provide a GO indication. If sync lock is not obtained, a NO-GO indication shall result. BIT step 11 shall also check the RF loop with the additional requirement that the 1.7 KHz test logic shall be exercised. In that case, the 30 dB attenuator switch in the transmitter output shall be activated, and the BIT signal strength shall decrease below the receiver threshold level. The SCP shall give a GO indication if phase lock is lost.

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BIT step 12 (loop test) shall route the RF signals through the forward antenna coaxial switch, A3K3, rather than the aft antenna coaxial switch, A3K4. BIT step 14 shall permit the SCP loop test circuitry to be tested alone by applying the BIT video output to the received video input connection and testing the video for sync lock with the generated BIT signal.

3.7.3.7.2 BIT panel. A BIT panel containing the BIT status and counter indicators and BIT sequencing switches shall be incorporated in the SCP front panel. The BIT RDY lamp shall indicate that the SCP +28 volts dc source is operating and the pod is ready for BIT. The AUTO START switch shall be a push-button switch and shall initiate automatic sequencing through the 15 test steps. Each test number shall be indicated by the correct combination of illuminated lamps 8, 4, 2, and 1. The GO lamp shall illuminate at the end of each successful test step. The NO-GO lamp shall illuminate and the test sequence shall automatically halt if a test is failed. The failed test step shall be indicated by the illuminated counter lamps. To proceed after a NO-GO step, the RESTART push-button switch shall be pressed. After step 15, either the GO or NO-GO lamp shall remain illuminated, indicating the condition of the SCP and the total pod. If a NO-GO occurred during any step sequence, then the NO-GO lamp shall remain on after step 15. The BIT shall be manually stepped through the test sequence by pressing the MANUAL START push-button switch for each test step.

3.7.4 Receiver, television R-2066A/AWW-7. The receiver, RE1, shall be a FM video receiver and shall meet the specifications of Drawing 196AS501-2.

3.7.4.1 Receiver functional description. Video data link signals, received at the selected antenna, shall be filtered by the diplexer and applied to the receiver. The receiver shall decode the received RF signal and provide a video signal output to the Signal Control-Processor. The video output from the receiver may be applied directly to the cockpit video monitor if necessary, bypassing the Signal Control-Processor.

3.7.4.2 Receiver input signal. The received video signal which modulates the RF carrier shall conform to MIL-T-81624. The receiver shall provide a valid receive signal output with an input of -87 dBm or higher input power and a minimum signal-to-noise ratio of 14 dB.

3.7.4.3 Receiver output video. The composite video output of the receiver shall be 3.0 volts peak-to-peak + 0.4 volts into 93 ohms resistance. The video shall have negative-going sync and shall conform to MIL-T-81624.

3.7.5 Tape Recorder RD-499A/AWW-7. The video tape recorder (VTR), A4, shall be a rugged, compact recorder/reproducer that will record the received video presentation during airborne conditions and play back the recorded information during ground operation maintaining clear and continuous quality video throughout the presentation. Simultaneously with the video recording, aircrew comments from the aircraft intercom system can be recorded on an audio track on the tape. The recorder shall be accessible for tape replacement and playback operation without requiring down loading of the pod from the aircraft.

3.7.5.1 Recorder operating environment. In general, the recorder shall operate under all service conditions to which the pod is subjected. The operating temperature range of the recorder, however, shall be limited to between -10^{0} and $+60^{0}$ Centigrade. The recorder shall receive no damage when operated in the standby mode, specified as recorder energized but not advancing tape, over the temperature range of -54^{0}C to $+71^{0}$ Centigrade.

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3.7.5.1.1 Recorder heater. The recorder shall contain a heating element controlled by a thermostatic switch which closes below 20°C. The temperature of the recorder, with covers secured, in the vicinity of the tape path and including the rotating video head drum, shall be maintained above 0°C for an ambient temperature down to -10°C. For a recorder initially at -10°C, the maximum warm-up time to 0°C shall be 20 minutes.

3.7.5.1.2 Recorder vibration isolation. The recorder shall be mounted on a base which is isolated from the pod casing assembly to reduce the effects of shock and vibration on the recording equipment. Four friction damped, vibration isolators, rated at 6 pounds force, and meeting the requirements of drawing 1035AS204, shall support the recorder, one at each corner, as it is mounted in the pod. Independent of the pod, the recorder, with shock mounts attached, shall operate as required when subjected to a vibration input of 0.01 g2/Hz from 10 to 2000 Hz in each of 3 mutually perpendicular planes. Sideways picture motion during vibration shall be acceptable, but there shall be no loss of sync.

3.7.5.2 Recorder physical dimensions. Maximum size of the recorder shall be not greater than 7 inches high, 12 inches wide, and 13 inches deep. Weight shall be not greater than 23 pounds (50.7 Kg) (including covers and connector). The recorder shall have a carriage handle which is not included in the above dimensions.

3.7.5.3 Recorder input/output connector. All electrical connections with the recorder shall be made through connector J1, part number M81511/01EB01P1. Connector J1 shall be located on the left side of the recorder as it is installed in the pod. Pin function assignment shall be as listed in table VI.

3.7.5.4 Recorder power. The recorder shall operate from +12 volts dc supplied by the signal control processor. The power requirement shall be 13 watts nominal. A heater installed in the recorder shall operate from +28 volts dc at power consumption of not greater than 52.5 watts.

3.7.5.5 Recorder controls. The recorder function and record controls, if required, shall be located behind a protective cover on the front side of the recorder. Upon application of power, the recorder shall be energized, but shall not advance tape. To advance tape and record a signal, +12 volts dc shall be applied to the record remote control pin in the connector. The recorder shall automatically shut off when all of the tape has been recorded.

3.7.5.6 Video recording system. The video recording system shall be a rotary, 2-head, helical scan system, using a full field, composite video signal based on American television standards. The recorder shall have a reel-to-reel tape system employing a 5-inch reel of video tape. The tape shall be chromium dioxide on polyester back and shall be 1 mil thick and 1/2 inch wide. The tape speed shall be 7-1/2 inches per second $\pm$ 0.25 percent. Recording time shall be not less than 27 minutes.

3.7.5.7 Recorder input video. The input signal shall be 1 volt peak-to-peak $\pm$ 0.5 volt composite video as measured into 75 ohm load. Sync shall be in the negative-going direction with respect to the video signal. The sync-only component of a signal shall have an amplitude of not less than 0.27 volts and not greater than 0.34 volts.

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TABLE VI. Pin function assignment.

Pin Number	Function
J1	
1	Video input
2	Video input shield
3	+28 volts dc to heater
4	Audio input shield
5	Video output
6	Video output shield
7	Ground
8	Audio input
9	Audio output shield
10	Audio output
11	Recorder remote control
12	+12 VDC power

3.7.5.8 Recorder output video. The output signal shall be 1 volt peak-to-peak, negative going sync, into a 75 ohm load. Resolution for the output video shall be not less than 5 shades of gray and 200 vertical lines. The signal-to-noise ratio shall be greater than 40 dB. The recorder shall have AGC (automatic gain control) such that a 10 dB decrease in input signal strength shall not adversely affect the video output signal.

3.7.5.9 Recorder audio. The audio signal level input to the recorder shall be 0.5 volts nominal into 30 k ohms. The playback audio output signal shall be 0.5 volts for 1 kHz into 1 k ohm. The frequency response shall be 100 to 10,000 Hz $\pm$ 3 dB. The signal-to-noise ratio shall be 40 dB.

3.7.5.10 Recorder BIT. The recorder shall contain BIT circuitry which shall monitor the following parameters:

a. 12 Vdc input power applied.

b. Video record mode circuits energized.

c. Recorder servo system sync in phase with the applied video sync.

The latter test shall verify that the recorder video head drum is rotating at the correct speed. LED indicators for each test parameter may be located on

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the front side of the recorder adjacent to the control panel and shall illuminate to provide a GO indication that the recorder is energized and ready to record input video signals.

3.7.5.11 Recorder reel stops. The recorder cover shall be secured in place prior to flight. Spacers mounted inside of the cover over the video tape reels shall secure the reels to the reel table and prevent the reels from coming off during vibration.

3.7.5.12 Recorder take-up reel tension. Provision shall be made to take up any slack which occurs in the tape path while the recorder is in the standby mode (recorder energized but not an advancing tape). To prevent tape from spilling off the supply reel the take-up reel shall recover slack tape and maintain alignment of the tape in the tape path through the recording mechanism.

3.7.5.13 Recorder marking. A tape threading diagram shall be located in the recorder cover. The recorder controls and functional positions shall be identified for the benefit of the operator.

3.7.6 Antenna-Radome AS-3220/AWW-7. The antenna-radome shall be manufactured in accordance with Drawing 196AS252 and shall meet the requirements of 3.7.6.1 through 3.7.6.3.

3.7.6.1 Antenna functional description. Either the forward antenna, E1, or the aft antenna, E2, may be selected for communication between the data link pod and weapon. The aft antenna shall provide continuous communication with the weapon after release and within the aft hemisphere of the pod. The fore antenna shall communicate with the weapon when it is located within the fore hemisphere of the pod. The selected antenna shall receive the video signals transmitted by the weapon and radiate telemetry signals to the weapon. The antenna shall be connected by coaxial cable to the diplexer in the casing assembly via the antenna select relay A3K2.

3.7.6.2 Antenna-radome physical description. The antenna-radome assembly shall consist of a fiberglass radome which contains a mechanically fixed, helical antenna. The helical antenna element shall project forward from a base plate which forms the antenna ground plane. The antenna base shall be secured to the radome bulkhead. The radome with the antenna mounted inside shall become an enclosed assembly preventing radio interference transmission or leakage into the pod casing assembly. The interior of the antenna-radome assembly shall be filled with a foam compound in accordance with Drawing 184AS290. The cured foam shall have a density of 2.00 $\pm$ 0.5 lbs/ft3 and a dielectric constant of 1.6 or less. Weight of the antenna-radome assembly shall be not greater than 24 pounds.

3.7.6.3 Antenna characteristics. The antenna shall use the nominal gain and beamwidth and maximum side lobes as specified for both the command and video RF frequencies in the contract. The antenna shall be circularly polarized with the main lobe on the longitudinal axis. The side lobes shall be down not less than 10 dB. The VSWR shall be not greater than 1.6:1 and the impedance shall be 50 ohms. The antenna shall contain an impedance matching device to meet this requirement.

3.7.7 Casing assembly. The casing assembly, A1, shall be manufactured in accordance with Drawing 875AS1005 and shall contain solid state electronic components, wiring cables, and associated mounting bases and hardware.

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3.7.7.1 Configuration. The body contour of the casing shall be continuous with the surface of the forward and aft antenna-radome assemblies where they are joined together. Internal arrangement and construction of the casing assembly shall permit removal and reinstallation of the component equipment contained therein through the access doors.

3.7.7.2 Access doors. Access to the components within the pod shall be accomplished through the two access doors in the casing assembly. The surface contour of the access doors, including fasteners and fastening hardware, shall be located within the major diameter of the casing assembly. The doors shall be sealed against moisture penetration around the opening and shall be electrically grounded to the casing to minimize transmission of electromagnetic radiation through the opening. Each door shall be marked with directions on how to open and close the door.

3.7.8 Filter assembly. The filter assembly, FL1, shall be manufactured in accordance with Drawing 68A25D62. It shall filter the aircraft ac power lines at the input to the pod to prevent broadband radio frequency interference from being conducted away from the equipment within the requirements of MIL-STD-461 for Class 1 equipment.

3.7.9 Breather system. A breather system meeting the specifications of Drawing 63A67A30 shall be installed in the lower aft end of the casing assembly.

3.7.10 Desiccant bags. Desiccant MIL-D-3464, type I, eight units, shall be enclosed in a retainer located at the forward end of the casing assembly to protect the video tape recorder from humidity and moisture condensation.

3.7.11 Wiring. Wiring shall meet the requirements of MIL-W-5088 except hook-up and harness wiring may be unmarked and uncoded.

3.7.12 Workmanship. Workmanship for the equipment shall conform to MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein all inspection shall be performed in accordance with the test conditions specified in 3.3.13 and 3.3.14.

4.4 First article inspection. First article inspection shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. First article inspection shall be accomplished under the approved test procedure of 4.5. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a Government representative may be designated to witness or supervise the tests when so desired.

4.4.1 Scope of tests. First article inspection shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. First article inspection shall include environmental tests in accordance with the procedures of MIL-T-5422 and interference tests in accordance with MIL-STD-462.

4.4.2 First article approval. Approval of the first article sample will be by the procuring activity upon satisfactory completion of all tests. tests. No production equipments shall be delivered prior to the approval of the first article sample. Prefabrication of production equipment prior to the approval of the first article sample is at the contractor's own risk. The approved first article sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The first article sample shall not be considered as one of the equipments under the contract.

4.4.3 Production equipments. Pods supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article sample. Each equipment shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection of a pod and of pods already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspection. All inspection and testing shall be under the supervision of the Government inspector. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests (see 4.5.1).

b. Sampling Tests (see 4.5.2).

c. Reliability Assurance Tests (see 4.5.3).

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d. Special Tests (see 4.5.4).

4.5.1 Individual tests. Each pod shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy, and reliability. As a minimum, each pod shall pass the following tests:

a. Examination of product (see 4.5.1.1).

b. Operational test (see 4.5.1.2).

c. Manufacturing run-in-test (see 4.5.1.3).

d. Workmanship vibration (see 4.5.1.4).

e. Evaluation of test (see 4.5.1.5).

f. Environmental burn-in-test (see 4.5.1.6).

4.5.1.1 Examination of product. Each pod should be examined carefully to determine that the pod complies with the requirements of 3.3.1, 3.3.2.4, 3.3.7.1, 3.3.7.2, 3.3.10, 3.3.11, 3.7.7, 3.7.11 and 3.7.12.

4.5.1.2 Operational test. Each pod shall be operated long enough to permit the pod temperature to stabilize and to check sufficient characteristics and record adequate data to assure required equipment operation.

4.5.1.3 Manufacturing run-in test. Each pod shall be operated under the conditions of workmanship vibration and environmental burn-in specified herein without failure. A failure shall be considered as anything which causes malfunctioning of the pod. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. Workmanship vibration shall precede the environmental burn-in test.

4.5.1.4 Workmanship vibration. Each pod shall be vibrated for a period of 15 minutes in the vertical plane using the shaped spectrum shown on figure 2. The pod shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Should a failure occur, it should be repaired and the test started over. The forward and aft doors may be removed from the pod in order to obtain access to the components and cables within the pod during vibration.

4.5.1.5 Evaluation of test. During vibration some snow interference and wavy motion on the video presentation is permissible. Tearing of the picture or black streaks in the video presentation shall be cause for rejection. The video strength shall not be affected during the vibration more than ± 1 dB referenced to the signal source. The video quality shall not be affected when command functions are energized. The transmitted command signal shall not be interrupted during the vibration exposure. Proper built-in-test (BIT) operation is not required during vibration; however, the BIT operation shall be normal after vibration. The video tape presentation can exhibit noise streaks in the video presentation, waving motion, and intermittent loss of sync during the vibration mode; however, a video tape presentation of video signals recorded after vibration shall appear normal.

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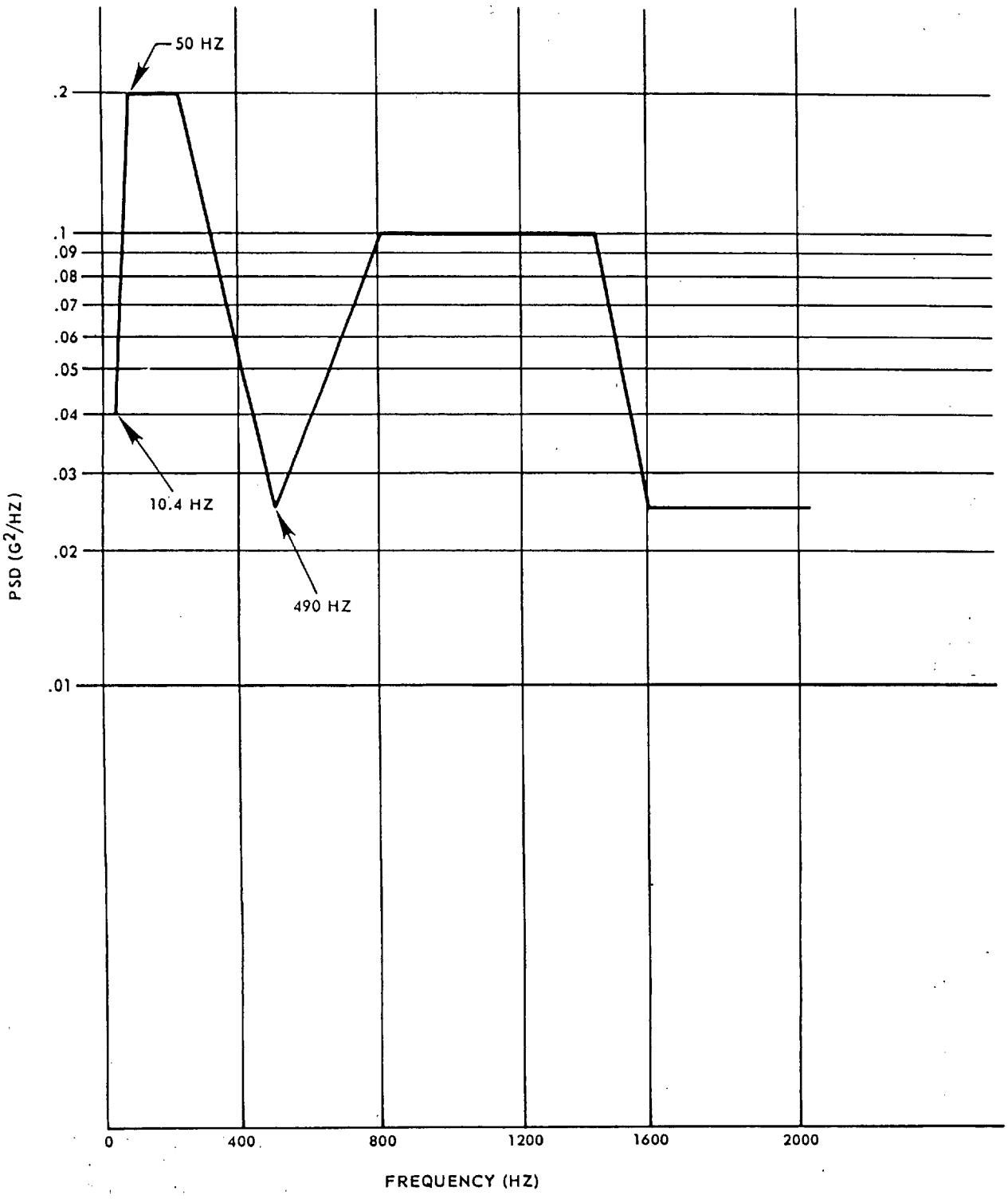


FIGURE 2. Random vibration power spectral density profile.

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4.5.1.6 Environmental burn-in-test. Each equipment shall be subjected to seven 12-hour environmental test cycles as shown on figure 3. Each test cycle shall consist of a 7-hour period of continuous pod operation preceded by a 5-hour cold temperature stabilization period. The test cycles shall include temperature cycling from -54° and low frequency sinusoidal vibration applied to the unit in test for 10 minutes per hour during the heating portion of each cycle. Vibration shall consist of a single non-resonant frequency between 20 and 60 Hz (measured at the mounting points on the equipment) having a peak amplitude of 2.2g ± 10 percent. The radomes may be removed from the equipment to accommodate the units in test to the test chamber. Each pod in test shall be subjected to an electrical performance test at +55°C as well as a brief performance check at -54°C once daily. At -54°C immediately following energization, each pod in test shall be monitored for proper video presentation on one of the five channels (1-5). A different channel shall be monitored each day such that all five channels are evaluated periodically during the test. Only pod operating time is counted as test time. Testing shall continue until each pod has successfully accumulated 49 hours of environmental test time, the last 14 of which must be failure free. The record of failures occurring during the environmental burn-in shall be maintained as the official record of the progress of the environmental burn-in test. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future pods. The environmental burn-in test shall not be required for pod that is subjected to the reliability assurance test.

4.5.2 Sampling tests. Pods selected for sampling tests shall first have passed the individual tests. Pods shall be selected for sampling tests by the Government inspector in accordance with table VII.

TABLE VII. Sampling test sample selection. 1/

Quantity of pods offered for acceptance	Quantity to be selected for sampling test
First 10	1
Next 50	1 for each additional 50 or fraction thereof

1/ Sampling tests are not required when reliability assurance tests are conducted.

4.5.2.1 Scope of tests. Each pod selected for sampling tests shall be subjected to the following tests:

a. Complete operational tests at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational tests at certain environmental conditions. The conditions may vary for each pod tested and shall be based on results of the preproduction, initial production, individual, and special tests.

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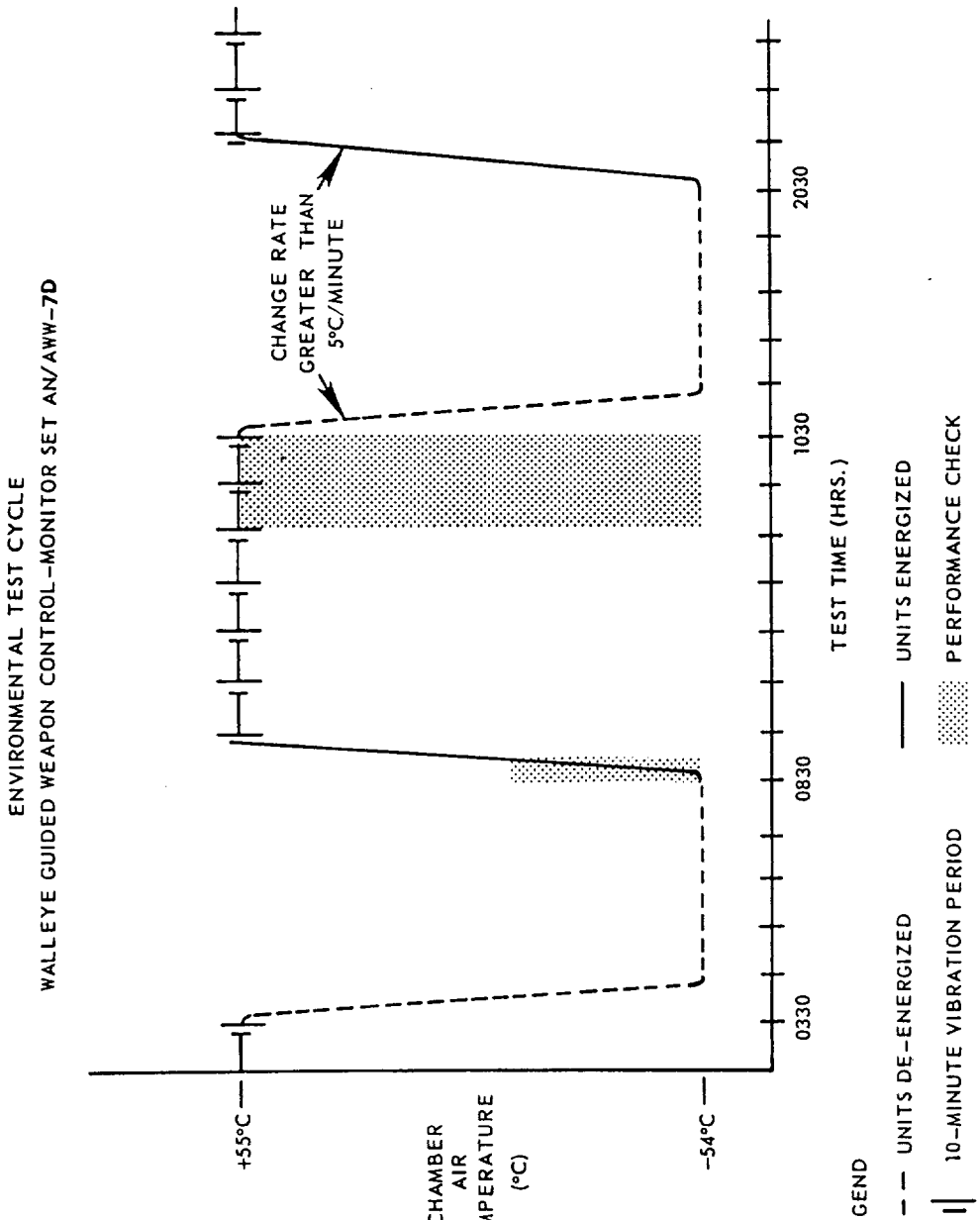


FIGURE 3. Reliability evaluation test

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c. Manufacturing run-in test specified in 4.5.1.3 except that the environmental burn-in test duration shall be 300 hours with no restriction on the number of failures.

4.5.3 Reliability assurance tests. Reliability assurance tests shall be conducted using MIL-STD-781. Tests as required by both the qualification phase and the production acceptance phase shall be conducted.

4.5.3.1 Qualification phase. Prior to the acceptance of pods under the contract or order, two of the first three production units shall be subjected to environmental cycling in accordance with MIL-STD-781, Test Level E, consisting of 12-hour environmental test cycles. Testing shall continue until a total of 1000 hours of operating time has been accumulated on the two units, or approximately 500 hours per pod. The radomes may be removed from the pod to accommodate the units in test to the test chamber.

4.5.3.1.1 Test procedure. Each test cycle shall consist of a 5-hour temperature stabilization period at -54 ^{o}C followed by a 7-hour period of continuous pod operation at +55 ^{o}C. Low frequency sinusoidal vibration shall be applied to the unit in test for 10 minutes each hour during the heating portion of each cycle. Input voltage cycling is not required. Each pod in test shall be subjected to an electrical performance test at +55 ^{o}C as well as a brief performance check at -54 ^{o}C once daily as shown on figure 3. At -54 ^{o}C immediately following energization, each pod in test shall be monitored for proper video presentation on one of the 5 channels (1-5). A different channel shall be monitored each day, such that all 5 channels are evaluated periodically during the test. Once weekly, each pod in test shall be subjected to a complete operational test at room ambient conditions. At the completion of the 1000 hours test time, each pod shall undergo a final operational test under standard conditions.

4.5.3.1.2 Evaluation of test. The total test time and total relevant failures accumulated during the reliability assurance test shall be used to determine compliance with the specified pod MTBF. Only pod operating time is counted as test time. Specifically, the test MTBF shall be calculated by dividing the total test hours by the number of relevant failures. If the test MTBF is equal to or greater than the specified MTBF for the pod and no pattern failures are present, the equipment is acceptable. If the test MTBF is lower, or if pattern failures exist, the pod shall be rejected and corrective action shall be accomplished. Reliability test reports on the qualification phase shall be prepared in accordance with MIL-STD-781.

4.5.3.2 Reliability acceptance test. Each pod produced, other than those submitted to the reliability assurance test, shall be submitted to the reliability acceptance test. The environmental burn-in test (see 4.5.16) shall be the basis from which accept/reject criteria for production reliability is determined.

4.5.3.2.1 Pre-test burn-in. A burn-in (or debugging) period prior to the test period of 4.5.1.6 may be used at the option of the contractor (see 6.2.1). If an additional burn-in period is to be used, the details thereof must be included in the approved test procedures. If used, the pre-test burn-in shall be the same for each pod in time and severity. Failures occurring during this period may be considered nonrelevant but shall be recorded.

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4.5.3.2.2 Evaluation of test. To determine whether the MTBF is being met at any time during the contract, the operating test hours and the failures thereon (not counting pre-test burn-in operating time and failures) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the reject line as necessary to accommodate the data; for example, see Reject Criteria for Test Plan XXIX in MIL-STD-781.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of pod at any time that a reject situation exists pending a review of the contractor's effort to improve the pods, the pod parts, or the equipment workmanship.

4.5.3.2.3 Disposition of tested equipment and report requirements. Each pod completing the test shall be submitted to a visual examination and final operational test under standard conditions. A final report on the reliability production acceptance phase shall be prepared.

4.5.3.3 Test details. The test details shall be part of the test procedures to be accepted by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests.

4.5.4 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the pod and to assure adequate quality control. The pod selected for special tests may be selected from pod previously subjected to the sampling or reliability assurance tests.

4.5.4.1 Special test schedule. Selection of pods for special tests shall be made as follows:

a. On an early pod after an engineering or material change.

b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.5.4.2 Scope of tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for acceptance prior to conducting the tests.

4.5.5 Pod failure. Should a failure occur during either the sampling, reliability assurance, or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or a design defect.

c. Where practical, include a test in the individual test to check all pods for this requirement until assurance is obtained that the defect has been corrected.

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4.5.6 Environment tests. The pod shall operate as required when subjected to the tests specified in 4.5.6.1 through 4.5.6.10.

4.5.6.1 Vibration. The pod shall meet the performance requirements of this specification during and after being subjected to the vibration environment specified herein.

4.5.6.1.1 Vibration test setup. The pod shall be mounted and tested with its longitudinal axis horizontal and positioned so that its mounting lugs are on top of the center body section. The pod shall be hand mounted directly to the shaker using a cradle to support the center body section. The stiffness of the mounting fixture shall be such that its induced resonant frequencies are as high as possible but none are less than 800 Hz.

4.5.6.1.2 Random vibration test. The pod shall be subjected to 4 hours of random vibration along each of the three mutually perpendicular axes of the pod. The power spectral density of the test control signal shall be 0.006 G^2/Hz over the frequency range from 10 to 1200 Hz and 0.022 G^2/Hz from 1201 to 2000 Hz.

4.5.6.1.3 Random vibration equalization. The accelerator spectral density applied to the equipment, as determined by an analyzing filter with a maximum bandwidth of 25 Hz between 20 and 200 Hz, shall be $\pm$ 2 dB over the frequency range from 10 Hz to 2000 Hz; all peaks and notches shall be within the $\pm$ 2 dB range. Tolerance levels in terms of dB are defined as follows:

$$dB = 10 \log \frac{G_1^2/Hz}{G_0^2/Hz} \quad \text{or} \quad dB = 20 \log \frac{G_1}{G_0}$$

where G_1^2 equals the acceleration power spectral density and G_1 equals Grms over the analyzer bandwidth. The term G_0, defines the specified level. (RMS levels shall be measured using a true RMS meter.)

4.5.6.2 Temperature-altitude. The pod shall be subjected to temperature-altitude combination requirements of MIL-STD-810, Method 520.0, except the altitude shall be not greater than 55,000 feet.

4.5.6.3 Humidity. The equipment shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507.2, Procedure I.

4.5.6.4 Shock. The pod shall be subjected to a shock test in accordance with MIL-STD-810, Method 516.3, Procedure I, except that the following deviations shall apply. One shock in each direction shall be applied along the vertical axis of the equipment (total of two shocks). The acceleration pulse shall be 20 G (sin t/.060) $\pm$ 1G. Tolerance limits of the half sine shock pulse configuration shall be in accordance with MIL-STD-810, Method 516.3, figure 516.3-3 where the peak value (A) is 20 G and the nominal duration (D) is 60 ms.

4.5.6.5 Sand and dust. The pod shall be subjected to sand and dust exposure as specified in MIL-STD-810, Method 510.2.

4.5.6.6 Fungus. The pod shall be subjected to fungus exposure as specified in MIL-STD-810, Method 508.3.

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4.5.6.7 Salt atmosphere. The pod shall be subjected to salt atmosphere exposure as specified in MIL-STD-810, Method 509.2.

4.5.6.8 Explosive conditions. The pod shall be subjected to explosive conditions as specified in MIL-STD-810, Method 511.2.

4.5.6.9 Rain. The pod shall be subjected to rainfall conditions equivalent to MIL-STD-810, Method 506.2, Procedure II. Covers and plugs shall be in place during exposure.

4.5.6.10 Acceleration. The pod shall be exposed to the design requirements of MIL-A-8591 for both wing and fuselage mounting. The pod shall withstand the aerodynamic and inertial captive flight loads specified in MIL-A-8591 for flight regimes up to MACH .95 at sea level.

4.6 Test procedures. The right is reserved by the procuring activity or the Government inspector to modify the test or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract.

4.7 Presubmission testing. No item, part or complete pod shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of this specification and the contract.

5. PACKAGING

5.1 General. The pod shall be packaged in a MK 426 Mod 0 shipping and storage container in accordance with MIL-E-17555, Preservation and Packaging Level A, Method IIa. The packaged pod shall be able to withstand storage, rehandling, and reshipment without necessity of repackaging.

5.2 Marking. Shipments shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The data link pod, Guided Weapon Control-Monitor Set AN/AWW-7D, is a major component part of the Walleye data link weapon system and provides two-way communication between a control aircraft and a guided weapon. The pod is designed for suspension from a standard Walleye configured bomb rack, and for non-jettisonable carriage on either the center line or inboard stores stations for a variety of high performance aircraft. Weapon control is accomplished from the cockpit via employment of a control unit and video monitor. The pod receives cockpit control signals and transmits them as encoded RF signals from the pod antenna to the weapon. Video signals from the weapon are received by the pod and processed for display on the cockpit video monitor. The data link pod will permit the control operator to monitor the flight of the weapon toward a target area and, if necessary, command the weapon to change its lock-on point after launch.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Procuring activity for placement of nameplate and identification markings (see 3.3.10).

c. Serial number for pod casing (see 3.3.10).

d. Oscillator frequency (see 3.7.2.9).

e. Transmitter and receiver carrier frequencies (see 3.4.1).

f. When a first article is required for inspection and approval (see 3.1, and 4.4).

g. When pre-test burn-in is used (see 4.5.3.2.1).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.2.1, 3.3	Engineering Change Proposal	DI-E-2037	---
3.3.6	Report, Electromagnetic Interference (EMI) Test	UDI-T-23723A	---
4.4	Procedure, First Article Inspection	DI-T-4901	---
4.4.2	Report, First Article Inspection	DI-T-4902	---
4.5.1.2	Report, Operational/ Performance Test	UDI-T-23740	---
4.5.1.6	Report, Burn-in Test	DI-R-7040	---
4.5.3.1.2	Report, Reliability and Maintainability Status	UDI-R-21137	---
4.5.3.2.1	Report, Failure	DI-R-21598	---
4.5.3.2.3	Report, Reliability Test and Demonstration	DI-RELI-80252	---

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4.5.3.3	Procedure, Reliability Test	DI-RELI-80251	---
4.5.4.2, 4.6	Production/Acceptance Inspection Procedures	DI-T-4903	---
4.5.5, 4.4.3	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719	---

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12-L, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.

6.3 First Article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Subject term (key word) listing.

Antenna-radome
Control-monitor set, guided weapon, AN/AWW-7D
Data link pod
Helical scan system
Signal control processor
Target tracking
Video recording system
Walleye guided weapon

Preparing Activity:
Navy-AS
(Project 1095-N237)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85392(AS)	Control-Monitor Set, Guided Weapon, AN/AWW-7D, General Specifica

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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